

- 1 **Military:** HMDs displays are built into fighter jet pilot's helmet to displays important information about the flight and target. Often they have useful features like night vision and the visors may even be protective. HMDs are also used for the army and police where the displays show vital information about the surroundings e.g. maps and thermal imaging data which can help detect a targets in very low light conditions. These HMDs also feature communication gear for coordination among soldiers as well as for receiving commands from base. Most military applications of HMD include a OLED display instead of a semi-reflective mirror.
- 2 **Sports:** Sports such as formula one racing can use AR with HMDs to show some important data about the race to the driver.

HMDs feature head movement detection thus providing an intuitive and immersive experience to the user by giving them six degrees of freedom

NOTE

Degrees of freedom is a concept of mechanical engineering which determines what type of motions are allowed for an object. The degrees are divided into two sets:

- 1 **Translational:** Movement of the object from one point to another. The three degrees under this consist of movements along the three axes of cartesian coordinates in space.
- 2 **Rotational:** Rotation of object in the directions pointed by the axes in the cartesian coordinate system.

Eye Glasses

Eye glasses that are used in AR are essentially stripped down versions of HMDs. They do not feature all the tactical advantage of HMDs and the augmentation is simpler. In most of the cases, eye glasses used for AR purposes look just like the normal glasses with an additional camera. These devices allow the user to see the world as they would wearing the normal glasses



AR eye glasses look like the normal ones, though with a camera

and add the augmentation onto the view. They feature OLED displays which allow the extra layer to be displayed directly on the glass. Some of them use the semi-reflective mirror approach as well.